

Auteur: Mohamed Shafik
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Bereken $\int \frac{x^4 dx}{\sqrt{x^5 - 6}}$

Substitutie: $t = x^5 - 6$, $dt = 5x^4 dx$, $x^4 dx = \frac{dt}{5}$

$$\begin{aligned}\int \frac{x^4 dx}{\sqrt{x^5 - 6}} &= \frac{1}{5} \int \frac{dt}{\sqrt{t}} \\ &= \frac{1}{5} \int t^{-1/2} dt \\ &= \frac{1}{5} \cdot 2t^{1/2} + c \\ &= \frac{2\sqrt{x^5 - 6}}{5} + c\end{aligned}$$

References